SUMAN MAITY PhD

ATMOSPHERIC SCIENTIST / GREENHOUSE-GAS INVERSION / CARBON–CLIMATE RESEARCH

Research Associate · Office for Integrated Emission Assessment, Earth System Division
National Institute for Environmental Studies (NIES) · Tsukuba, Japan

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PROFESSIONAL PROFILE

Atmospheric scientist working at the intersection of global greenhouse-gas transport, observation-constrained inverse modelling and climate–carbon-cycle variability. Expertise spans satellite and in-situ CO_2 observations, NICAM-TM/4D-Var inversion systems, and regional climate modelling. Research addresses the variability of terrestrial carbon uptake and Indian summer monsoon processes.

SELECTED SCIENTIFIC CONTRIBUTIONS

- Led a 2026 Geophysical Research Letters study showing that weakened northern mid-to-high-latitude land CO_2 uptake contributed 0.42 PgC yr^{-1} to the 2023–2024 non-fossil flux anomaly, alongside tropical ENSO-related changes.
- Released the NISMON- CO_2 observation-constrained global surface-flux dataset (2010–2025), generated using GOSAT X CO_2 and NICAM-TM/4D-Var inversion.
- Led NISMON-CO_2 inversion flux using GOSAT XCO_2 in Global Carbon Budget 2026.
- Participated in OCO2 V11 MIP and contributed NISMON-CO_2 inversion flux using OCO2, OCO3 datasets.
- Conducted CO_2 transport/inversion and ensemble data-assimilation research; earlier studies examined monsoon variability, convection and land-surface sensitivities using regional climate models.

ACADEMIC & RESEARCH APPOINTMENTS

Research Associate | National Institute for Environmental Studies, Japan — *May 2024–present*

NICAM-TM, adjoint/4D-Var CO_2 inversions, GOSAT XCO_2 assimilation and carbon-cycle diagnostics.

Postdoctoral Researcher | JAMSTEC, Japan — *Oct 2021–Mar 2024*

MIROC4-ACTM atmospheric transport and LETKF-based greenhouse-gas assimilation.

Associate Researcher | Sun Yat-sen University, China — *Jan 2020–Jan 2021*

Research Scientist | IIT Bombay, India — *Feb–Dec 2019*

Faculty of Mathematics | Bankura Unnayani Institute of Engineering, India — *Mar 2006–Jul 2010*

Faculty of Mathematics | College of Engineering and Management, India — *Jun–Dec 2005*

EDUCATION

PhD, Atmospheric Modelling | Indian Institute of Technology Kharagpur, India | 2018

Thesis: Investigation of interannual and intraseasonal variability of Indian Summer Monsoon using high resolution regional climate model.

MSc, Applied Mathematics | Vidyasagar University, India

BSc (Honours), Mathematics | Vidyasagar University, India

RESEARCH METHODS & TECHNICAL EXPERTISE

Models and inversion NICAM-TM, NISMON-CO_2, MIROC4-ACTM, RegCM4, WRF; 4D-Var, adjoint, LETKF, inverse modelling.

Computing and analysis Python (xarray, NumPy, pandas, Matplotlib, Cartopy), Fortran, Bash, CDO, NetCDF/HDF5, MPI, Linux and HPC/Fugaku.

SELECTED PEER-REVIEWED PUBLICATIONS

Reverse chronological selection · DOI links are clickable · Full publication list: Google Scholar

- Maity, S., Niwa, Y., Saeki, T., Someya, Y., & Yoshida, Y. (2026). **Drivers of 2023–2024 atmospheric CO_2 growth: Role of northern mid-to-high latitude land carbon cycle.** *Geophysical Research Letters, 53(13), e2026GL122221*. DOI
- Singh, K. S., Nayak, S., Maity, S., Nayak, H. P., & Dutta, S. (2023). **Prediction of extremely severe cyclonic storm "Fani" using moving nested domain.** *Atmosphere, 14(4), 637*. DOI
- Maity, S., Nayak, S., Nayak, H. P., & Bhatla, R. (2022). **Comprehensive assessment of RegCM4 towards interannual variability of Indian summer monsoon using multi-year simulations.** *Theoretical and Applied Climatology, 148, 491–516*. DOI
- Patra, P. K., Dlugokencky, E. J., Elkins, J. W., … Maity, S., & Takigawa, M. (2022). **Forward and inverse modelling of atmospheric nitrous oxide using MIROC4-atmospheric chemistry-transport model.** *Journal of the Meteorological Society of Japan, 100(2), 361–386*. DOI
- Nayak, S., Takemi, T., & Maity, S. (2022). **Precipitation and temperature climatologies over India: A study with AGCM large ensemble climate simulations.** *Atmosphere, 13(5), 671*. DOI
- Maity, S., Nayak, S., Singh, K. S., Nayak, H. P., & Dutta, S. (2021). **Impact of soil moisture initialization in the simulation of Indian summer monsoon using RegCM4.** *Atmosphere, 12(9), 1148*. DOI
- Nayak, S., Mandal, M., & Maity, S. (2021). **Assessing the impact of land-use and land-cover changes on the climate over India using a regional climate model (RegCM4).** *Climate Research, 85, 1–20*. DOI
- Cheng, J., Li, Q., Chao, L., Maity, S., Huang, B., & Jones, P. (2020). **Development of high resolution and homogenized gridded land surface air temperature data: A case study over Pan-East Asia.** *Frontiers in Environmental Science, 8, 588570*. DOI
- Nayak, S., Mandal, M., & Maity, S. (2018). **RegCM4 simulation with AVHRR land use data towards temperature and precipitation climatology over Indian region.** *Atmospheric Research, 214, 163–173*. DOI
- Maity, S., Satyanarayana, A. N. V., Mandal, M., & Nayak, S. (2017). **Performance evaluation of land surface models and cumulus convection schemes in the simulation of Indian summer monsoon using a regional climate model.** *Atmospheric Research, 197, 21–41*. DOI
- Maity, S., Mandal, M., Nayak, S., & Bhatla, R. (2017). **Performance of cumulus parameterization schemes in the simulation of Indian summer monsoon using RegCM4.** *Atmósfera, 30(4), 287–309*. DOI

OPEN RESEARCH DATA

Maity, S. (2026). **Global CO_2 fluxes derived from NISMON-CO_2 and satellite XCO_2.** NIES Global Environmental Database; monthly global fluxes, 2010–2025. Dataset DOI

SELECTED CONFERENCE PRESENTATIONS

- IWGGMS-22 (2026): How sensitive is a CO_2 flux inversion to GOSAT-family XCO_2 retrievals?
- AOGS (2026): Seasonal climate driven CO_2 flux variability in northern mid-to-high latitudes.
- IWGGMS-21 (2025): 2023–2024 anomalous CO_2 growth using GOSAT X CO_2-constrained inverse modelling.
- EGU General Assembly (2025): Global CO_2 flux estimation using NISMON- CO_2 and GOSAT.
- Japan Meteorological Society Autumn Meeting (2025): Inverse modelling of the 2023–2024 CO_2 surge.